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OPTI-X GALAXY PG

AERO NEXUS

PG Business Analytics Game 2026 — Full Technical & Functional Documentation

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1. Project Overview

1.1 Introduction

OPTI-X GALAXY PG (codenamed *AERO NEXUS*) is a real-time, multiplayer business analytics simulation game designed for postgraduate students. Built for the **Analytics Konclave 2026** at **KIIT School of Management (KSOM)**, it challenges teams to forecast airline seat occupancy by analyzing 60 historical monthly records and predicting Economy and Premium Economy seat demand across 6 competitive rounds.

1.2 Game Concept

Teams take on the role of **Business Analytics Consultants** supporting the demand-planning team of a growing Indian airline. The airline operates a fleet of narrow-body aircraft on domestic trunk routes and needs accurate seat-demand forecasts for both Economy and Premium Economy cabins. Forecasts feed into the Revenue Management System (RMS) — over-forecasting leads to surplus unsold inventory; under-forecasting means rejected bookings and revenue leakage.

1.3 Key Features

Feature	Description
60 Historical Monthly Records	Rich training dataset with 15 demand driver variables
6 Live Rounds	Each round presents a unique monthly scenario (Monsoon, Festive, Wedding, etc.)
Real-Time Timer	Configurable countdown timer (default 10 minutes per round)
Dual Forecasting	Teams predict Economy Seats (~16,000–22,000) and Premium Economy Seats (~1,600–2,500)
Weighted Scoring	60% Economy accuracy + 40% Premium accuracy, with penalty modifiers
Live Leaderboard	Real-time competitive rankings
Admin Command Center	Full game master control panel
Session Conflict Handling	Prevents multi-device login with force-login option
Bulk Team Import	Excel-based team onboarding
Export Capabilities	Export teams and submissions to Excel

2. Tech Stack

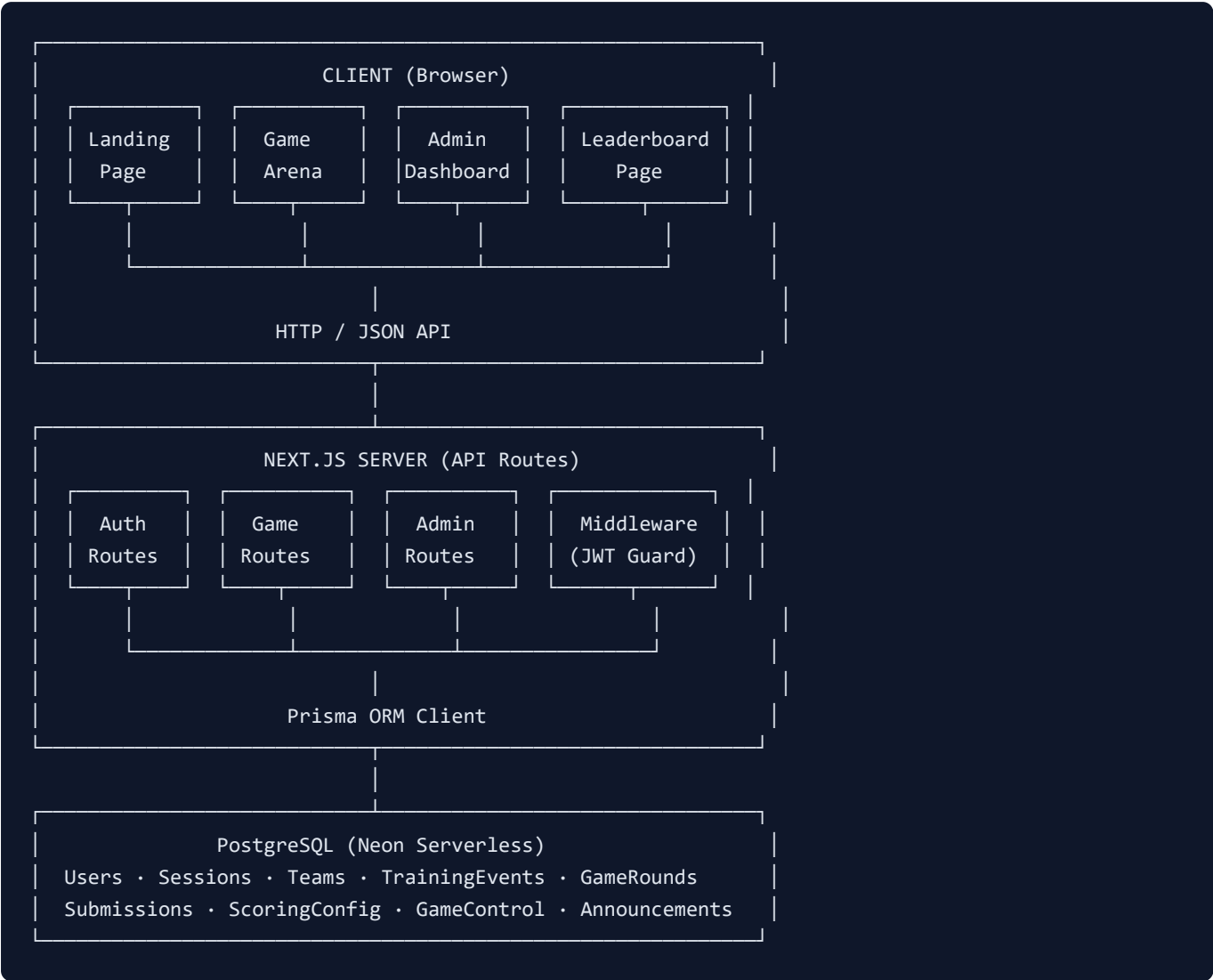
2.1 Core Technologies

Layer	Technology	Version
Framework	Next.js (App Router)	14.2.35
Language	TypeScript	5.x
UI Library	React	18.x
Styling	Tailwind CSS	3.x
Animations	Framer Motion	—
Icons	Lucide React	—
Charts	Recharts	—
Database	PostgreSQL (Neon Serverless)	—
ORM	Prisma	6.19.3
Auth (JWT)	jose	—
Password Hashing	bcryptjs	—
Excel I/O	xlsx (SheetJS)	—

2.2 Development Tools

Tool	Purpose
ESLint	Code linting
PostCSS	CSS processing pipeline
Prisma Studio	Database GUI (<code>npm run db:studio</code>)

3. Architecture Overview



4. Database Schema

4.1 Entity Relationship Overview

The database consists of **10 core models** managed via Prisma ORM.

4.2 Model Definitions

4.2.1 User (Admin Accounts)

Field	Type	Description
id	String (cuid)	Primary key
email	String (unique)	Admin email address
passwordHash	String	bcrypt-hashed password
role	String (default: "ADMIN")	User role
createdAt	DateTime	Account creation timestamp

4.2.2 Session

Field	Type	Description
id	String (cuid)	Primary key
userId	String	FK → User
tokenHash	String	Hashed session token
expiresAt	DateTime	Session expiry
isActive	Boolean	Active session flag

4.2.3 Team (Player Teams)

Field	Type	Description
id	String (cuid)	Primary key
teamCode	String (unique)	Team identifier (e.g., "TEAM-01")
name	String	Display name
pin	String	bcrypt-hashed access PIN
rawPin	String?	Plaintext PIN (for admin visibility)
currentRound	Int (default: 1)	Current round the team is on
totalScore	Float (default: 0)	Cumulative score
status	String (default: "active")	Team status
institution	String?	Institution name
sessionToken	String?	Active session token
sessionStartedAt	DateTime?	When the current session started
roundStartedAt	DateTime?	When the current round timer started
createdAt	DateTime	Registration timestamp

Relations: submissions → Submission[]

4.2.4 TrainingEvent (60 Historical Monthly Records)

Field	Type	Description
id	Int (autoincrement)	Primary key
eventId	String (unique)	Event identifier (T001–T060)
t	Int	Time period index
monthOfYear	Int	Month (1–12)
isSummerVacation	Int (0/1)	Summer vacation flag
isMonsoon	Int (0/1)	Monsoon season flag
isFestiveHoliday	Int (0/1)	Festive/holiday flag
isWeddingSeason	Int (0/1)	Wedding season flag
fuelPriceIndex	Float	Aviation fuel price index
leisureDemandIndex	Float	Leisure travel demand index
corporateActivityIndex	Float	Corporate travel activity index
capacityEconomy	Int	Available economy seats capacity
capacityPremium	Int	Available premium seats capacity
competitorCapacityIndex	Int	Competitor airlines capacity index
avgFareEconomy	Float	Average economy fare (₹)
avgFarePremium	Float	Average premium fare (₹)
promoIntensity	Int	Promotional intensity level
weatherDisruptionIndex	Float	Weather disruption severity index
specialEventFlag	Int (0/1)	Special event flag (elections, IPL, etc.)
economySeats	Int	Actual economy seats sold (answer)
premiumEconomySeats	Int	Actual premium economy seats sold (answer)

4.2.5 GameRound (6 Live Rounds)

Field	Type	Description
id	Int (autoincrement)	Primary key
roundNumber	Int (unique)	Round 1–6
title	String	Round title (e.g., "Monsoon Disruption")
theme	String	Thematic scenario
instructions	String	Detailed scenario narrative
All 15 demand variables	Various	Same variables as TrainingEvent
actualGa	Int	Correct Economy Seats answer
actualVip	Int	Correct Premium Economy Seats answer
status	String (default: "ready")	Round status
timerSeconds	Int (default: 600)	Timer duration per round

4.2.6 The 6 Round Scenarios

Round	Title	Month	Key Theme	Economy Answer	Premium Answer
R1	Monsoon Disruption	July (7)	Heavy monsoon flooding, weather delays	16,450	1,720
R2	Fuel Duty Hike + Independence Day	Aug (8)	Fuel cost spike + holiday travel bump	18,650	1,920
R3	GST Rate Cut	Sep (9)	Tax reduction passed to passengers	19,850	2,050
R4	Diwali Festive Peak	Oct (10)	Year's strongest leisure/home-travel peak	21,200	2,280
R5	Wedding Season + Corporate Year-End	Nov (11)	Peak wedding + corporate travel	21,650	2,450
R6	New Year Holidays	Dec (12)	Christmas/New Year travel, cautious corporate	21,100	2,320

4.2.7 Submission

Field	Type	Description
id	String (cuid)	Primary key
teamId	String	FK → Team
roundNumber	Int	Which round (1–6)
predictedGa	Int	Team's Economy Seats prediction
predictedVip	Int	Team's Premium Economy prediction
reasoning	String?	Team's analytical reasoning
gaError	Float	Absolute error for Economy
vipError	Float	Absolute error for Premium
accuracy	Float	Combined accuracy score
score	Float	Final round score
submittedAt	DateTime	Submission timestamp

Constraints: Unique on [teamId, roundNumber]

4.2.8 ScoringConfig

Field	Type	Default	Description
gaWeight	Float	0.6	Weight for Economy accuracy (60%)
vipWeight	Float	0.4	Weight for Premium accuracy (40%)
maxRoundScore	Float	1000	Maximum score per round
overForecastPenalty	Float	1.0	Penalty multiplier for over-forecasting
underForecastPenalty	Float	1.0	Penalty multiplier for under-forecasting

4.2.9 GameControl

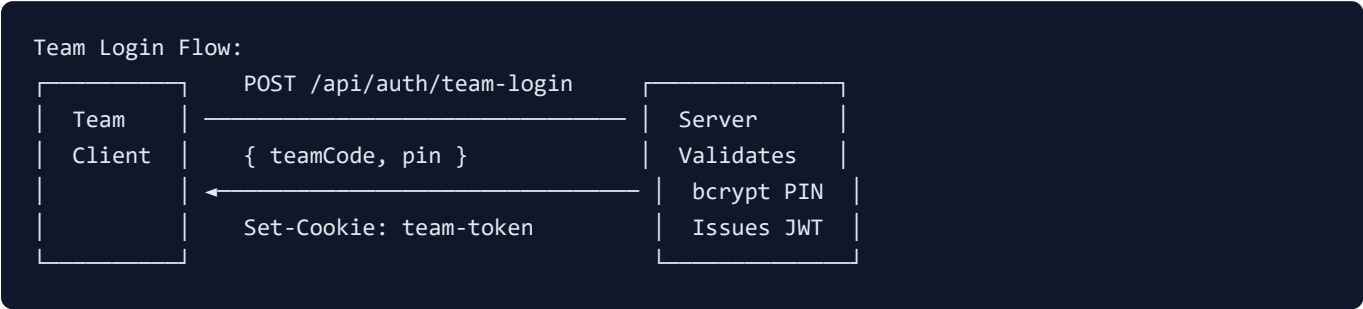
Field	Type	Default	Description
<code>gameStatus</code>	String	"lobby"	Current game state
<code>activeRound</code>	Int	1	Active round number
<code>timerEnabled</code>	Boolean	true	Whether timer is active
<code>leaderboardEnabled</code>	Boolean	true	Whether leaderboard is visible
<code>timerStartedAt</code>	DateTime?	—	When the timer was started
<code>roundTimerSeconds</code>	Int	600	Timer duration

4.2.10 Announcement & AuditLog

Standard models for system-wide broadcast messages and action logging respectively.

5. Authentication & Authorization

5.1 Team Authentication



- **Endpoint:** `POST /api/auth/team-login`
- **Input:** `teamCode` (string) + `pin` (string)
- **Validation:** PIN verified against bcrypt hash in database
- **Session:** Sets `team-token` HTTP-only cookie (12-hour expiry)
- **Conflict Handling:** If a team is already logged in elsewhere, returns `SESSION_CONFLICT` error. Client can send `forceLogin: true` to override.

5.2 Admin Authentication

- **Endpoint:** `POST /api/auth/login`
- **Input:** `email` (string) + `password` (string)
- **Validation:** Password verified against bcrypt hash
- **Session:** Generates a JWT via `jose` containing `{ userId, role: 'ADMIN' }`, stored in `oxg-session` HTTP-only cookie (24-hour expiry)

5.3 Middleware Protection

The middleware at `src/middleware.ts` intercepts all requests:

Route Pattern	Protection
<code>/game/*</code>	Requires valid <code>team-token</code> cookie
<code>/admin/*</code>	Requires valid <code>oxg-session</code> JWT with <code>role === 'ADMIN'</code>
All other routes	Public access

6. API Reference

6.1 Authentication Endpoints

POST /api/auth/login

Purpose: Admin login

Body: { email: string, password: string }

Response: { success: true } + sets oxg-session cookie

Errors: 401 Invalid credentials

POST /api/auth/team-login

Purpose: Team login

Body: { teamCode: string, pin: string, forceLogin?: boolean }

Response: { success: true, teamId, teamCode, name } + sets team-token cookie

Errors: 401 Invalid credentials, 409 SESSION_CONFLICT, 403 GAME_COMPLETED

POST /api/auth/logout / POST /api/auth/team-logout

Purpose: Clear session cookies

Response: { success: true }

GET /api/auth/me

Purpose: Get current authenticated user/team info

Response: { type: 'admin'|'team', data: {...} }

POST /api/auth/register

Purpose: Register new team or admin account

Body: { teamCode, name, pin, institution? } or { email, password }

6.2 Game Endpoints

GET /api/game/status

Purpose: Get current game state

Auth: team-token

Response:

```
{
  "currentRound": 1,
  "status": "active",
  "timerState": {
    "timerStarted": true,
    "remaining": 540,
    "total": 600
  }
}
```

GET /api/game/round/[number]

Purpose: Fetch round scenario data

Auth: team-token

Response: Round variables (month, fuel index, demand indices, capacity, fares, etc.) without answers

GET /api/game/training-data

Purpose: Fetch all 60 historical training events

Auth: team-token

Response: Array of TrainingEvent objects for analysis

POST /api/game/submit

Purpose: Submit team predictions

Auth: team-token

Body:

```
{
  "roundNumber": 1,
  "predictedGa": 16500,
  "predictedVip": 1750,
  "reasoning": "Heavy monsoon + high weather disruption index of 8 suggests reduced leisure demand..."
}
```

Response: Submission result with score breakdown

POST /api/game/start-round

Purpose: Start the round timer for the team

Auth: team-token

Response: { started: true, timerSeconds: 600 }

GET /api/game/results

Purpose: Fetch team's past submission results

Auth: team-token

Response: Array of Submission objects with scores

GET /api/game/leaderboard

Purpose: Get ranked team standings

Response: [{ teamCode, name, totalScore, rank }]

GET /api/game/announcements

Purpose: Get active broadcast messages

Response: Array of Announcement objects

6.3 Admin Endpoints

GET/PUT /api/admin/game-control

Purpose: Read/update global game state

Auth: oxg-session (ADMIN)

PUT Body: { gameStatus, activeRound, timerEnabled, leaderboardEnabled }

GET/PUT /api/admin/scoring

Purpose: Read/update scoring configuration

Auth: oxg-session (ADMIN)

PUT Body: { gaWeight, vipWeight, maxRoundScore, penalties }

GET/POST /api/admin/teams

Purpose: List teams / Create new team

Auth: oxg-session (ADMIN)

POST /api/admin/bulk-import

Purpose: Import teams from Excel file

Auth: oxg-session (ADMIN)

Body: FormData with .xlsx file

GET /api/admin/rounds

Purpose: List all 6 game rounds with answers

Auth: oxg-session (ADMIN)

GET /api/admin/answers

Purpose: View correct answers for all rounds

Auth: oxg-session (ADMIN)

GET /api/admin/submissions

Purpose: View all team submissions across rounds

Auth: oxg-session (ADMIN)

POST /api/admin/announcement

Purpose: Create a new broadcast announcement

Auth: oxg-session (ADMIN)

GET /api/admin/game/export/teams

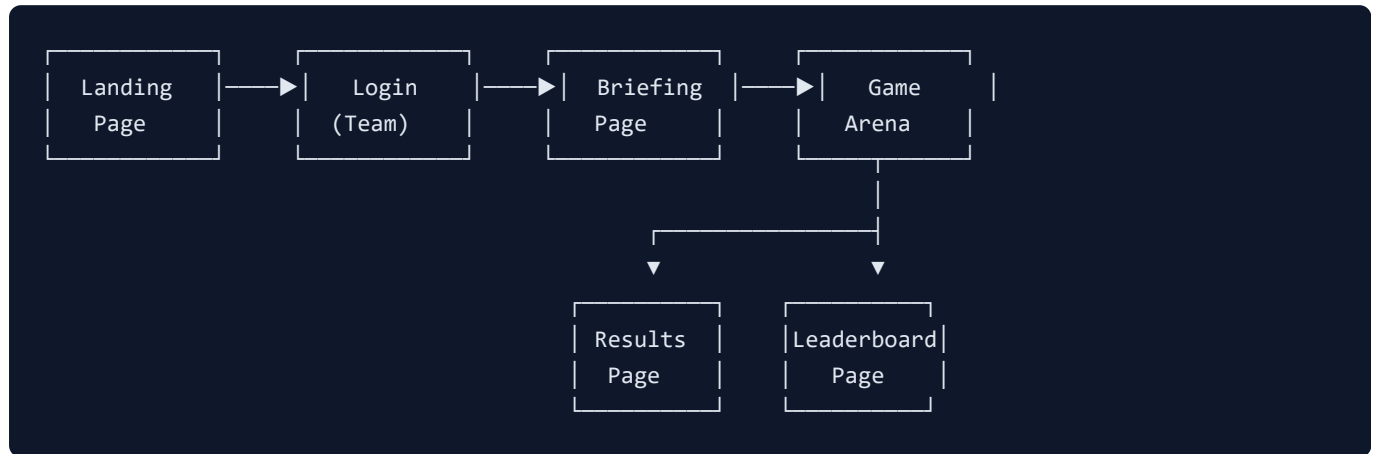
GET /api/admin/game/export/submissions

Purpose: Export data as Excel downloads

Auth: oxg-session (ADMIN)

7. Frontend Pages & User Flow

7.1 User Journey



7.2 Page Details

Landing Page (/)

- Hero section with airplane background image and "AERO NEXUS" title
- Badge: "PG Business Analytics Game 2026"
- Quick stats: 60 Months · 6 Rounds · 10m/Round · 100 Max Pts
- Login form with Team Code + Access Code (with eye toggle)
- Session conflict handling with force login option
- Game completion "Thank You" screen
- Overview section: "The Challenge at a Glance" (4 steps)
- Variable dictionary section listing all 15 demand drivers
- Footer with KSOM & Analytics Konclave branding

Team Login (/login)

- Standalone login card with Team Name + Access Code
- Eye toggle for PIN visibility
- Session conflict handling
- Link to Admin Portal

Admin Login (/login/admin)

- Admin branding with Shield icon and "Admin Access" badge
- Email + Password fields with eye toggle
- "ACCESS COMMAND CENTER" button

Case Study (/case-study)

- Full business scenario narrative
- Airline context and revenue management background

Guidelines (/guidelines)

- Game rules and participation guidelines

Game Briefing (/game/briefing)

- War Room Briefing with case study
- Tabbed interface: Case Study · Round Brief · Data & Resources
- Business scenario: Airline demand-planning consultants
- Forecasting targets: Economy (16,000–22,000) and Premium (1,600–2,500)
- Variable dictionary with 15 demand drivers (seasonal flags, indices, capacity, fares)
- Data download buttons (Excel/CSV for 60 historical records)
- Round scenario reveal with all input variables and narrative

Game Arena (/game)

- Active round display with countdown timer
- Round scenario card with all 15 variables
- Dual input fields: Economy Seats (0–25,000) and Premium Economy Seats (0–5,000)
- Reasoning text area for analytical justification
- Submit button with confirmation modal
- Phase management: loading → briefing → prep → playing → review → confirm → result → transition → completed

Market Analysis (/game/market)

- Interactive charts via Recharts
- All 60 historical monthly records displayed
- Trend visualization for fuel, demand, fare, and occupancy

Results (/game/results)

- Round-by-round score breakdown
- Economy and Premium accuracy percentages
- Cumulative score tracking

Leaderboard (/game/leaderboard)

- Ranked team standings
 - Total score display
 - Real-time updates
-

8. Components Library

8.1 Shared Components

LogoHeader.tsx

- Displays KSOM logo + divider + Analytics Konclave logo
- Center title: "AERO NEXUS"
- Subtitle: "PG Business Analytics Game 2026"

Navbar.tsx

- Main navigation bar with responsive design
- Team info display

ThemeToggle.tsx

- Light/Dark mode switcher
- Persists user preference

CaseStudyModal.tsx

- Full-screen modal overlay
- Contains complete airline scenario, rules, and variable explanations

8.2 Game Components

ResourcePanel.tsx

- Complex data visualization panel using Recharts
 - Renders 60 historical training events as interactive charts
 - Trend analysis for fuel prices, demand indices, fares, capacity, and occupancy
 - Used during gameplay for pattern analysis and forecasting reference
-

9. Game Mechanics & Scoring

9.1 Game Structure

Phase	Description	Duration
Lobby	Teams register and wait	Until admin starts
Round 1–6	Each round presents a unique monthly scenario	10 min (default)
Completed	All rounds finished, final rankings shown	—

9.2 The 15 Demand Driver Variables

#	Variable	Type	Range/Values	Description
1	Month of Year	Int	1–12	Calendar month
2	Summer Vacation	Binary	0/1	Summer vacation period flag
3	Monsoon	Binary	0/1	Monsoon season flag
4	Festive Holiday	Binary	0/1	Festive holiday flag (Diwali, Independence Day, etc.)
5	Wedding Season	Binary	0/1	Peak wedding season flag
6	Fuel Price Index	Float	~85–110	Aviation turbine fuel price index
7	Leisure Demand Index	Float	~90–150	Leisure travel demand indicator
8	Corporate Activity Index	Float	~95–135	Corporate/business travel indicator
9	Capacity Economy	Int	~17,000–22,000	Available economy seats
10	Capacity Premium	Int	~1,800–2,500	Available premium economy seats
11	Competitor Capacity Index	Int	~95–115	Competing airlines' capacity
12	Avg Fare Economy	Float	~₹2,800–4,300	Average economy ticket fare
13	Avg Fare Premium	Float	~₹7,500–10,800	Average premium ticket fare
14	Promo Intensity	Int	0–3	Promotional campaign intensity
15	Weather Disruption Index	Float	0–10	Severity of weather disruptions
16	Special Event Flag	Binary	0/1	National events (elections, IPL, etc.)

9.3 Forecasting Targets

Target	Range	Description
Economy Seats	~16,000 – 22,000	Economy cabin seats sold
Premium Economy Seats	~1,600 – 2,500	Premium economy cabin seats sold

9.4 Scoring Algorithm

The scoring engine is implemented in `src/lib/scoring.ts`:

Score Calculation:

1. Economy Error = $|\text{predictedEco} - \text{actualEco}| / \text{actualEco} \times 100$
2. Premium Error = $|\text{predictedPrem} - \text{actualPrem}| / \text{actualPrem} \times 100$
3. Economy Accuracy = $\max(0, 100 - \text{Economy Error})$
4. Premium Accuracy = $\max(0, 100 - \text{Premium Error})$
5. Weighted Accuracy = $(\text{Economy Accuracy} \times \text{gaWeight}) + (\text{Premium Accuracy} \times \text{vipWeight})$
Default weights: `gaWeight = 0.6` (Economy), `vipWeight = 0.4` (Premium)
6. Penalty Adjustments:
 - Over-forecast: multiply error by `overForecastPenalty` (default 1.0)
 - Under-forecast: multiply error by `underForecastPenalty` (default 1.0)
7. Final Score = $(\text{Weighted Accuracy} / 100) \times \text{maxRoundScore}$
Default `maxRoundScore = 1000`

9.5 Timer Mechanism

- Timer is managed globally via `GameControl.timerEnabled` and per-team via `Team.roundStartedAt`
- When a team starts a round, `roundStartedAt` is set to the current timestamp
- Remaining time = `timerSeconds - elapsed seconds since roundStartedAt`
- Timer state is persisted in `localStorage` for same-device recovery
- Server is authoritative — client syncs from `/api/game/status`

10. Admin Dashboard

10.1 Dashboard Overview (/admin)

- Game status summary
- Active round indicator
- Total teams count
- Submission statistics

10.2 Admin Pages

Page	Path	Features
Overview	/admin	Dashboard stats, quick actions
Teams	/admin/teams	List all teams, view PINs (eye toggle), create/delete teams, generate random PINs
Bulk Import	/admin/teams/bulk-import	Upload Excel file to create multiple teams
Game Control	/admin/game-control	Set active round, enable/disable timer, toggle leaderboard, control game state
Rounds	/admin/rounds	View all 6 rounds with scenario details and answers
Scoring	/admin/scoring	Configure weights, penalties, max scores
Submissions	/admin/submissions	View all team submissions across all rounds
Answers	/admin/answers	View/reveal correct answers for each round
Announcements	/admin/announcements	Create broadcast messages for teams
Leaderboard	/admin/leaderboard	Admin view of rankings

10.3 Team Management Features

- **Create Team:** Manual creation with custom team code and PIN
- **Generate PIN:** Auto-generate random secure PINs
- **Eye Toggle:** Show/hide PINs in the team table
- **Bulk Import:** Upload .xlsx file with team data

- **Export:** Download team data as Excel
 - **Session Management:** View active sessions, force logout
-

11. Types & Utilities

11.1 Core Types (`src/types/index.ts`)

```
interface RoundData {
  roundNumber: number;
  title: string;
  theme: string;
  instructions: string;
  monthOfYear: number;
  isSummerVacation: number;
  isMonsoon: number;
  isFestiveHoliday: number;
  isWeddingSeason: number;
  fuelPriceIndex: number;
  leisureDemandIndex: number;
  corporateActivityIndex: number;
  capacityEconomy: number;
  capacityPremium: number;
  competitorCapacityIndex: number;
  avgFareEconomy: number;
  avgFarePremium: number;
  promoIntensity: number;
  weatherDisruptionIndex: number;
  specialEventFlag: number;
}

interface SubmissionResult {
  predictedGa: number;
  predictedVip: number;
  gaError: number;
  vipError: number;
  accuracy: number;
  score: number;
}

interface TrainingEvent {
  eventId: string;
  t: number;
  monthOfYear: number;
  // ... all 15 demand driver variables
  economySeats: number;
  premiumEconomySeats: number;
}
```

ROUND_THEMES Constant

Defines thematic scenarios for each of the 6 rounds: 1. Monsoon Disruption 2. Fuel Duty Hike + Independence Day 3. GST Rate Cut 4. Diwali Festive Peak 5. Wedding Season + Corporate Year-End 6. New Year Holidays

11.2 Utility Functions (`src/lib/`)

`auth.ts`

- `createToken(payload)` — Creates JWT via jose
- `verifyToken(token)` — Verifies JWT and returns payload
- `hashPassword(password)` — bcrypt hash (12 salt rounds)
- `comparePassword(input, hash)` — bcrypt compare
- `getTeamFromCookie()` — Extracts team info from cookie
- `getAdminFromCookie()` — Extracts admin info from cookie

`scoring.ts`

- `calculateScore(predicted, actual, config)` — Core scoring algorithm

`utils.ts`

- `cn(...classes)` — Tailwind class merger (clsx)
- `formatNumber(n)` — Number formatting with commas
- `formatTime(seconds)` — MM:SS timer display
- `getRelativeTime(date)` — "2 minutes ago" style formatting

`prisma.ts`

- Global Prisma client singleton (prevents connection exhaustion in dev)
-

12. Configuration & Environment

12.1 Environment Variables

Variable	Required	Description
DATABASE_URL	✓	PostgreSQL connection string (pooled)
DIRECT_URL	✓	Direct database connection (for migrations)
JWT_SECRET	✓	Secret key for JWT signing
NEXT_PUBLIC_APP_URL	✗	Public application URL
NODE_ENV	✗	Environment mode (development/production)

12.2 Build Configuration

next.config.mjs : - ESLint errors ignored during build (ignoreDuringBuilds: true) - TypeScript errors ignored during build (ignoreBuildErrors: true)

tsconfig.json : - Path alias: @/* → ./src/* - Strict mode enabled

13. Deployment Guide

13.1 Prerequisites

- Nodejs 18+
- PostgreSQL database (Neon recommended)
- npm or yarn

13.2 Local Development

```
# 1. Clone repository
git clone https://github.com/Avinash-Panda-gg/optix-pg-game.git

# 2. Install dependencies
npm install

# 3. Configure environment
cp .env.example .env
# Edit .env with your database credentials

# 4. Initialize database
npm run db:push    # Push schema to database
npm run db:seed    # Seed with training data + admin account

# 5. Start development server
npm run dev        # Runs on http://localhost:3000
```

13.3 Production Build

```
npm run build    # Runs: prisma generate && next build
npm start        # Starts production server
```

13.4 Vercel Deployment

1. Connect GitHub repo to Vercel
2. Set environment variables in Vercel dashboard
3. Build command: `npx prisma generate && next build`
4. After deploy: Run `npx prisma db push` and `npx prisma db seed`

13.5 Default Admin Credentials

Field	Value
Email	admin@pg-game.com
Password	Admin@123

 **Change default credentials immediately after first login.**

14. Project Structure

```

opti-x-galaxy-pg/
├── prisma/
│   ├── schema.prisma          # Database schema definition
│   └── seed.ts                 # Database seed script (60 events + 6 rounds)
├── public/
│   └── airplane.png           # Hero background image
├── src/
│   ├── app/
│   │   ├── api/
│   │   │   ├── auth/
│   │   │   │   ├── login/route.ts
│   │   │   │   ├── team-login/route.ts
│   │   │   │   ├── logout/route.ts
│   │   │   │   ├── team-logout/route.ts
│   │   │   │   ├── register/route.ts
│   │   │   │   └── me/route.ts
│   │   │   ├── game/
│   │   │   │   ├── status/route.ts
│   │   │   │   ├── round/[number]/route.ts
│   │   │   │   ├── training-data/route.ts
│   │   │   │   ├── submit/route.ts
│   │   │   │   ├── start-round/route.ts
│   │   │   │   ├── results/route.ts
│   │   │   │   ├── leaderboard/route.ts
│   │   │   │   └── announcements/route.ts
│   │   │   └── admin/
│   │   │       ├── teams/route.ts
│   │   │       ├── bulk-import/route.ts
│   │   │       ├── game-control/route.ts
│   │   │       ├── scoring/route.ts
│   │   │       ├── rounds/route.ts
│   │   │       ├── answers/route.ts
│   │   │       ├── submissions/route.ts
│   │   │       ├── announcement/route.ts
│   │   │       └── game/export/
│   │   │           ├── teams/route.ts
│   │   │           └── submissions/route.ts
│   │   ├── admin/
│   │   │   ├── layout.tsx
│   │   │   ├── page.tsx
│   │   │   ├── teams/
│   │   │   │   ├── page.tsx
│   │   │   │   └── bulk-import/page.tsx
│   │   │   ├── game-control/page.tsx
│   │   │   ├── rounds/page.tsx
│   │   │   ├── scoring/page.tsx
│   │   │   ├── submissions/page.tsx
│   │   │   ├── answers/page.tsx
│   │   │   ├── announcements/page.tsx
│   │   │   └── leaderboard/page.tsx
│   │   └── game/
│   │       ├── layout.tsx
│   │       └── page.tsx

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```
├── briefing/page.tsx
├── market/page.tsx
├── results/page.tsx
├── leaderboard/page.tsx
├── login/
│   ├── page.tsx
│   └── admin/page.tsx
├── register/page.tsx
├── case-study/page.tsx
├── guidelines/page.tsx
├── layout.tsx
├── page.tsx
├── components/
│   ├── shared/
│   │   ├── LogoHeader.tsx
│   │   ├── Navbar.tsx
│   │   ├── ThemeToggle.tsx
│   │   └── CaseStudyModal.tsx
│   └── game/
│       └── ResourcePanel.tsx
├── lib/
│   ├── auth.ts
│   ├── scoring.ts
│   ├── utils.ts
│   └── prisma.ts
├── types/
│   └── index.ts
├── middleware.ts
├── .env.example
├── next.config.mjs
├── tailwind.config.ts
├── tsconfig.json
├── package.json
└── README.md
```

OPTI-X GALAXY PG — AERO NEXUS

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